

ETSI/DECT-2020 Component RIT Technology Implementation and Evaluation Contributions in mMTC and URLLC Use Cases

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Abstract—We propose a comprehensive evaluation of ETSI/DECT-2020 New Radio (NR) as an IMT-2020 candidate technology for the mMTC (connection density) and URLLC (reliability) use cases. Several independent evaluation groups assessed DECT-2020 NR performance against the IMT-2020 requirements and INRS played a key role within the Canadian Evaluation Group (CEG) in evaluating the technology via simulation. In order to develop 5G link- and system-level simulators, it is necessary to identify and understand all key characteristics of DECT technology. Nevertheless, the mMTC use case offers D2D communications, which made simulation analysis extremely cumbersome. In this paper, we also explain the different maneuvers in order to fasten the simulations and reduce complexity while respecting all DECT specifications. Simulation results provided from link- and system-level simulations for both URLLC and mMTC use cases show that DECT-2020 component RIT meets the IMT-2020 requirements.

Index Terms—5G; ETSI/DECT; Technology evaluation; Connection density (mMTC); Reliability (URLLC); D2D; channel mismatch; implementation impairments; channels estimation errors; synchronization; localization; direction-of-arrival (DoA)

I. INTRODUCTION

The International Telecommunication Union Radiocommunication sector (ITU-R) has set the requirements for International Mobile Telecommunications-2020 (IMT-2020 5G Standard). With the rise of new wireless communications, artificial intelligence, and big data, several enabling radio technologies standardized by different standardization organisms such as 3GPP and ETSI have been proposed within the ITU-R Working Party 5D.

The European telecommunications standards Institute (ETSI)-digital enhanced cordless telecommunications (DECT) Forum was submitted to ITU-R; claiming to have met the minimum technology requirements that need to be fulfilled in order to pass [1]. Independent evaluation groups (IEG) participated in evaluating DECT submission involving Ultra Reliable Low Latency Communications (URLLC) and massive Machine Type Communications (mMTC) and INRS took part of the process within the Canadian evaluation group (CEG) using guidelines for the evaluation process in [2]. The DECT evaluation also garnered the attention of the research community and led to several evaluation publications [3]–[6].

The DECT-2020 NR offers star and mesh network topologies to support URLLC and mMTC use cases, respectively. In early

submissions, ETSI TC DECT did not provide all technical specifications for mMTC use case, thereby preventing from deciding whether the technology fulfills the specifications or not. After a new submission [7]–[10] and several technical exchanges, it was possible to run link- and system-level simulations to evaluate connection density. DECT-2020 NR has to fulfill the requirement of 99.999% reliability and support more than one million devices per km^2 .

In this paper, we point out key features of DECT-2020 NR and most importantly the D2D challenge in our simulations and the maneuvers made to the routing process to make such complex simulations in terms of time and memory run smoothly. We propose a comprehensive evaluation of ETSI/DECT-2020 New Radio (NR) as an IMT-2020 candidate technology for the mMTC (connection density) and URLLC (reliability). Several independent evaluation groups assessed DECT-2020 NR performance against the IMT-2020 requirements and INRS played an important part within the Canadian Evaluation Group (CEG) in evaluating the technology via simulation.

In order to develop 5G link- and system-level simulators, it is necessary to identify and understand all key characteristics of DECT technology. Nevertheless, the mMTC use case offers D2D communications, which made simulation analysis extremely cumbersome. [11] analyzed by system level simulations the energy consumption of DECT-2020 NR networks with different network sizes and topologies. In this paper, we also explain the different maneuvers in order to fasten the simulations and reduce complexity while respecting all DECT specifications.

Simulation results provided from link- and system-level simulations for both URLLC and mMTC use cases show that DECT-2020 component RIT meets the IMT-2020 requirements.

The terminology (traditionally used in DECT) Fixed Part (FP) and Portable Part (PP) is used to identify the two DECT radio end points. The FP is normally fixed and corresponds to what is referred to as BS in mobile terminology, the PP is normally mobile and corresponds to what is referred to as UE in mobile terminology.

The paper is organized as follows. Section II describes the specifications. Assumptions and configuration for DECT evaluation are detailed in Section III. Numerical evaluation results are analyzed in Sections IV and V draws out concluding

TABLE I
EVALUATION PARAMETERS FOR UMI-STREET CANYON AND UMA SCENARIOS [12].

Parameters	UMi - street canyon	UMa
Cell layout	Hexagonal grid, 19 micro sites, 3 sectors per site (ISD = 200m)	Hexagonal grid, 19 macro sites, 3 sectors per site (ISD = 500m)
BS antenna height h_{BS}	10m	25m
UT location: indoor/outdoor	indoor and outdoor	indoor and outdoor
UT location: LOS/NLOS	LOS and NLOS	LOS and NLOS
Indoor UT ratio	80%	80%
UT mobility (horizontal plane only)	3km/h	3km/h
Min. BS - UT distance (2D)	10m	35m
UT distribution (horizontal)	uniform	uniform

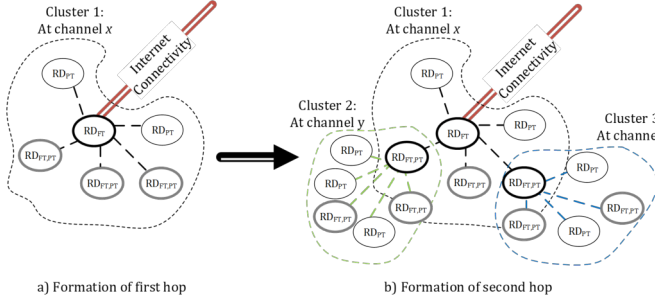


Fig. 1. Formation of the clustered tree mesh network topology [7].

remarks.

II. DECT SPECIFICATIONS

DECT-2020 NR has to fulfill the requirement of 99.999% reliability and support more than one million devices per km^2 [1]. These requirements are set by ITU-R WP5D to support different 5G use cases. To do so, DECT-2020 NR offers a new physical layer numerology, advanced medium access control algorithms, and supports mesh network topology in addition to the star topology [9].

DECT-2020 NR technical specifications show that any Radio Device (RD) in the network is able to transmit and receive are free to choose any operation mode simultaneously or not simultaneously (fixed termination mode or portable termination mode). A RD operating in fixed mode provides information to other RDs on how to initiate a connection and communicate with it and is also responsible for routing the data.

As seen in Fig. 1, the mesh network topology is realized by adding more RDs that connect themselves to any of the network nodes and autonomously choose or change their role between routing and non-routing, thereby, offering a scalable solution for mMTC use case.

In the mMTC environment, several assumptions are made, which include high tower and high transmission power macro cellular base station network deployment. Key DECT features such as gateways instead of high-power base stations are adopted in DECT to operate in the DECT band.

Each gateway has three independent sinks that operate as DECT FP. All devices have capability to route other devices data. Urban Street canyon model was adopted as the D2D channel and its specifications were taken from (3GPP TR

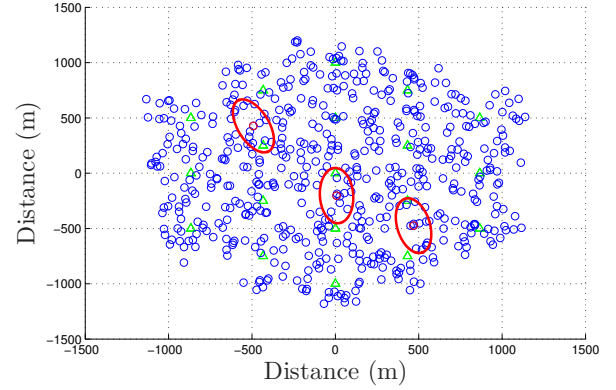


Fig. 2. D2D relaying UEs selection process.

38.901 V16.1.0) [12] where mMTC devices deployed in low heights and it has higher shadow fading compared to e.g. Urban Macro. Some key specifications can be shown in Table I. Additionally, it is assumed that all links between mMTC devices to be NLOS to avoid too positive pathloss values between mMTC devices.

Regarding the mesh connection topology, all nodes can send Beacon with very low rate. In multi channel case, they select used channel based on interference detect in each channel i.e. other beacons and can measure the RSSI and list all detected beacons where the RSSI value is high with reasonable detection quality (where there is a specific threshold sensitivity for D2D communication). Thus, nodes that are too far away are excluded. Nodes select next communication hop to the node sending the beacon that minimizes number hops to BTS. Each communication link is decode-and-forward with link specific (H)-ARQ to ensure that transmission is not lost. In simulation, fixed MSC was used. All transmissions are QPSK with coding.

III. ASSUMPTIONS AND CONFIGURATION FOR DECT EVALUATION

Table II enumerates the parameters used for connection density evaluation in mMTC use case while Table III presents the parameters for reliability evaluation in URLLC use case.

We had to tackle multiple issues to produce the desired DECT evaluation metric. Full specifications required to set up

TABLE II
ASSUMPTIONS AND CONFIGURATION OF URBAN MACRO-MMTC.

Parameters	DECT	INRS
Communication direction	Uplink	Uplink
Channel bandwidth	1.728 MHz	1.728 MHz
Number of channels	1 and 3	5
Traffic model	Poisson; 1 packet/2h/device; non-full buffer model	Poisson; 1 packet/2h/device; non-full buffer model
RD deployment	80% indoor, 20% outdoor	80% indoor, 20% outdoor
Number of FTs	19	19
Number of RDs	4M	4M
Proportion of RD-Fs	N/A	Dynamic
Maximum number of hops	Up to 255	Up to 255
Inter-site distance (ISD)	500 m	500 m
RX sensitivity	99.7 dBm	99.7 dBm
Transmission power	23 dBm	23 dBm
Thermal noise power	174 dBm/Hz	174 dBm/Hz
Noise figure	7 dB	7 dB
Carrier frequency	700 MHz	700 MHz
Layer 2 packet size	32 bytes	32 bytes
Slot length	417 us	417 us
Routing	Minimum hops	Minimum hops
Channel access	Random access, listen before talk	Random access, listen before talk
Number of random access slots	FT: 9598; RD-F: 46	FT: 9598; RD-F: 46
Number of antenna elements	RD: 1; FT: 1	RD: 1; FT: 1
Maximum antenna gains	RD: 0 dBi; FT: 8 dBi	RD: 0 dBi; FT: 8 dBi
Antenna heights	RD: 1.5 m; FT: 25 m	RD: 1.5 m; FT: 25 m
Spatial diversity	None	None
Path loss model (incl. shadowing)	UMa-B, 3GPP-D2D	UMa-B, 3GPP-D2D
Link fading model	UMa-B, 3GPP-D2D	UMa-B, 3GPP-D2D
Subcarrier spacing	27 kHz	27 kHz
FFT length	64	64
Modulation and coding scheme	MCS Index 1	MCS Index 1
Maximum HARQ retransmissions	3	1
Channel estimation	N/A	MMSE
RD velocity	3 km/h	3 km/h

the simulator were not entirely confirmed until after the 37th WP5D meeting. Besides, given the final setup adopted and translating into exhaustive consideration a priori of all potential D2D channels between the user equipment (UE)/devices and all potential multi-hop relaying paths (up to 255 in number), the amount of data required to be generated for around 77,000 UE/devices per sector/cell to reflect the criterion of 1 million users per km² was cumbersome. Just a single drop to carry out a simulation could take days for one BS site and, as a consequent extension, many weeks for the full layout of nineteen BS sites. Despite having powerful computing servers, simply obtaining the required memory allocations was not possible. After the 38th meeting of WP5D in June 2021, during which the DECT-2020 proponent provided the necessary clarifications, significant progress was made to successfully develop and integrate in the system-level simulator several processing-acceleration schemes that ultimately overcame the overwhelming a) memory consumption required by the DECT submission and b) computational complexity.

Other evaluation groups, added additional simplification parameters such as maximum number of hops and proportion of RD-Fs where one can choose just a set of RDs capable of routing data while in the specifications, all RDs are able to route. Our team proposed to adopt a dynamic approach to route data

without generating all D2D channel coefficients between all RDs. In fact, as shown in Fig. 2, when a certain RD is not able to establish a communication, it starts searching for a potential relay from a set of potential RDs drawn from an elliptical zone defined between the RD and the base station. This zone covers a small area and dynamically changes until finding a relay that satisfies the specification defined by DECT-2020. In doing so, only some D2D channel coefficients are generated dynamically and simulation complexity reduces drastically without fixing a specific set of relaying nodes. Once the relay is chosen, the number of hops increases by 1 and the same process is again applied if 1 hop is not enough to establish the communication. This technique is not only much simpler but also adapts with each environment to find the set of relays able to establish a communication while reducing considerably the simulation complexity. The proposed simplification let the evaluation happen while respecting the configuration assumptions without affecting the system performance.

IV. NUMERICAL EVALUATION RESULTS

The Wireless Lab developed both link- and system-level simulators using Matlab in order to evaluate both mMTC and URLLC environments for ETSI/DECT candidate. In the first phase, simulations were carried on without adopting the proposed strategy (i.e., each RD is capable of routing data and

TABLE III
ASSUMPTIONS AND CONFIGURATION OF URBAN MACRO-URLLC (DOWNLINK CASE).

Parameters	ITU-R M.2412	INRS
Baseline evaluation configuration parameters		
System architecture	Star with 19 sites, each site has 3 TRxPs (cells). Device connects directly to FT.	Star with 19 sites, each site has 3 TRxPs (cells). Device connects directly to FT.
Carrier frequency for evaluation	700MHz	700MHz
Channel model	Urban Macro	Urban Macro
BS antenna height	25m	25m
Total Tx Power per TRxP in BS/sink	49dBm for 20MHz	Results provided with 23dBm and 49dBm for 20MHz
UE/node power class	23	Not applicable as DL only
Percentage of high loss and low loss building type	100% low loss	100% low loss
Percentage of high loss and low loss building type	100% low loss	100% low loss
Additional parameters for system-level simulation		
Inter-site distance	500m	500m
Number of antenna elements per TxRP	Up to 64	Results provided with 20, (5x4)
UE antennas	Up to 4	4
Device deployment	80% indoor, 20% outdoor	80% indoor, 20% outdoor
Inter-site interference modelling	Explicitly modelled	Explicitly modelled
BS noise figure	5 dB	Not applicable DL only
UE noise figure	7 dB	7 dB
BS/sink antenna element gain	8 dBi	8 dBi
UE antenna element gain	0 dBi	40 dBi
Thermal noise level	174 dBm/Hz	174 dBm/Hz
Traffic model	Full buffer	Full buffer
Physical layer packet size	Not defined	Not applicable but Single 32 byte can be packet mapped to long pre-amble packet modulated according MCS2 (QPSK), in link simulations.
Simulation bandwidth	Up to 100 MHz	Channel Bandwidth 1.728 MHz.
UE density	10 UEs per TRxP	10 UEs per TRxP
UE antenna height	1.5m	1.5m

trying all possible combinations is compulsory). This approach did not work as expected since the computational cost is cumbersome. Applying in a second phase the proposed dynamic selection strategy to our simulator made the production of the following results possible owing to tremendous computational gains. Therefore, for mMTC use case, 5 channels were used with 19 FTs and 1 million RDs per km^2 , which translates to more than 4 million RDs in a configuration of 19FTs. As explained in section III, we did not set a fixed proportion of RD-Fs for data relaying but the number is dynamic and adaptative to each environment without limiting the number of maximum hops. The RX sensitivity for routing is set to 99.7 dBm with a routing cost that aims to minimize the number of hops and a maximum HARQ retransmissions of 1. Channel access, antennas configuration and other specifications are taken from DECT [7]–[10].

To evaluate the performance of the mMTC case, we measured the Packet Error Rate (PER) at the receiver. Table IV explains the two approaches following ITU M. 2412 [2] and DECT's methodology which calculates the packet outage rate instead of connection density. The requirement is fulfilled if the packet outage rate is lower than 1% for UEs density of 1 000 000 devices per km^2 .

By conducting Mont-Carlo experiments, the PER plotted in Fig. 3 curves show the same behavior as those obtained by

DECT's self-evaluation. Table VI shows that the packet outage is 0.2138% for a threshold of 1%, which validates that DECT-2020 component RIT meets the IMT-2020 connection density requirements.

For URLLC use case, star architecture is adopted (mesh architecture for mMTC use case) with devices connecting directly to FTs. The number of antenna elements per TxRP is set to 20 (5x4) while UEs antennas is set to 4. All other specifications are taken from DECT documents and some of them are presented in Table III.

From Table V, we make similar conclusions for reliability requirement in the URLLC use case. Indeed, DECT-2020 component RIT meets the IMT-2020 reliability requirements. For further details regarding simulation results and interpretations, please refer to CEG report [13].

We produced for the CEG ten confidential evaluation reports representing the official results, both provisional and final, that cover not only ETSI/DECT, but also 3GPP and NuFront [13]–[15].

V. CONCLUSIONS

In this paper, we proposed a comprehensive evaluation of ETSI/DECT-2020 New Radio (NR) as an IMT-2020 candidate technology for the mMTC (connection density) and URLLC (reliability) use cases. Several independent evaluation groups assessed DECT-2020 NR performance against the IMT-2020

TABLE IV
MMTC SIMULATION APPROACH FOLLOWING THE ITU M.2412 GUIDELINES AND DECT APPROACH.

DECT non-full buffer system-level simulation	ITU non-full buffer system-level simulation [2]	ITU full buffer system-level simulation followed by link-level simulation [2]
- Step 1: Set system user number 1 000 000 devices per km^2. - Step 2: Generate the user packet according to the traffic model. - Step 3: Run non-full buffer system-level simulation to obtain the packet outage rate. The outage rate is defined as the ratio of the number of packets that failed to be delivered to the destination receiver within a transmission delay of less than or equal to 10s to the total number of packets generated in Step 2. - Step 4: Increase the traffic density. - Step 5: Calculate the packet outage rate. The requirement is fulfilled if the packet outage rate is lower than 1%.	- Step 1: Set system user number per TRxP as N. - Step 2: Generate the user packet according to the traffic model. - Step 3: Run non-full buffer system-level simulation to obtain the packet outage rate. The outage rate is defined as the ratio of the number of packets that failed to be delivered to the destination receiver within a transmission delay of less than or equal to 10s to the total number of packets generated in Step 2. - Step 4: Change the value of N and repeat Step 2-3 to obtain the system user number per TRxP N satisfying the packet outage rate of 1%. Step 5: Calculate connection density. The requirement is fulfilled if the connection density C is greater than or equal to 1 000 000 devices per km^2.	- Step 1: Perform full-buffer system-level simulation using the evaluation parameters for Urban Macro-mMTC test environment, determine the uplink $SINR_i$ for each percentile $i=199$ of the distribution over users, and record the average allocated user bandwidth. - Step 2: Perform link-level simulation and determine the achievable user data rate R_i for the recorded $SINR_i$ and W_{user} values, according to the traffic model. - Step 3: Calculate the packet transmission delay of a user. Step 4: Calculate the traffic generated per user. - Step 5: Calculate the long-term frequency resource requested under $SINR_i$. - Step 6: Calculate the number of supported connections per TRxP. - Step 7: Calculate the connection density. The requirement is fulfilled if the 99th percentile of the delay per user D_i is less than or equal to 10s, and the connection density is greater than or equal to 1 000 000 devices per km^2.

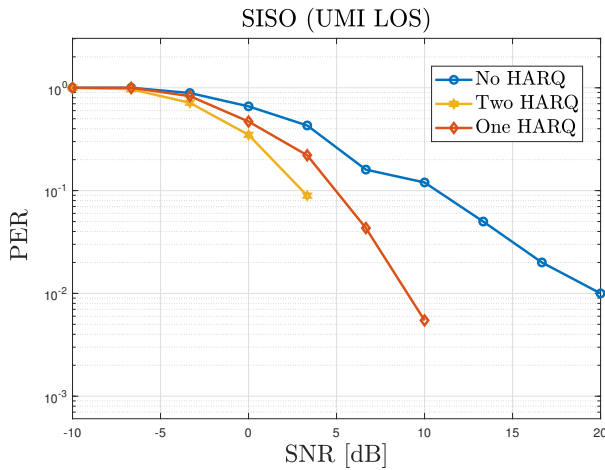


Fig. 3. PER versus SNR (SISO).

TABLE V
URBAN MACRO-URLLC (CONFIGURATION A CHANNEL MODEL A) - FDD.

Metric	Link	M.2410	INRS
Reliability [%]	DL	99.999%	>99.999%
Reliability [%]	UL	99.999%	>99.999%

TABLE VI
URBAN MACRO-mMTC (CANYON STREET D2D CHANNEL MODEL).

Metric	Link	DECT (threshold)	INRS
Packet outage [%]	UL	1%	0.2138%

requirements and INRS played an key role within the Canadian Evaluation Group (CEG) in evaluating the technology via simulation. In order to develop 5G link- and system-level simulators, it is necessary to identify and understand all key characteristics of DECT technology. Nevertheless, the mMTC use case offers D2D communications, which made simulation analysis extremely cumbersome. In this paper, we also explain the different maneuvers in order to fasten the simulations and reduce complexity while respecting all DECT specifications. We showed that generating the D2D channels in a dynamic way using ellipses in certain keys areas makes simulations a lot faster and feasible. Simulation results provided from link- and system-level simulations for both URLLC and mMTC use cases show that DECT-2020 component RIT meets the IMT-2020 requirements.

Furthermore, some future work directions consist not only in achieving a more comprehensive evaluation of ETSI/DECT a technology (i.e., instead of evaluating the minimum requirements that are sufficient for the candidate to pass), but also in improving the proposed algorithmic implementations of the simulators to further reduce the overwhelming computational cost of any link- and system-level simulations involving a D2D communication layer while maintaining optimal performance.

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